

November 13, 2025

Master Degree in Bionics Engineering

Course of Principles of Bionics and Biorobotics Engineering

Lesson title:

Elements of a Robot

Type of robots

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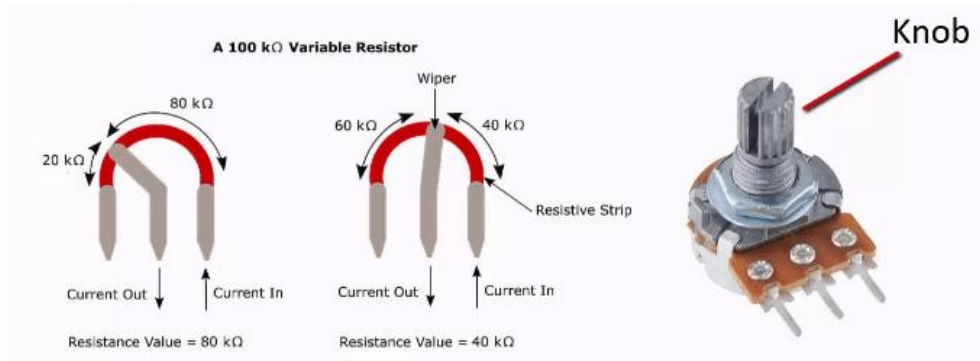


Elements of a Robot

Sensors

Proprioceptive Sensors

Measure the robot's internal states, such as **joint or wheel position and speed**.



Potentiometer



Encoder

Elements of a Robot

Sensors

Exteroceptive Sensors

Measure **environmental variables** around the robot, such as temperature, distance to obstacles, object position/orientation, or forces exchanged with the environment.

Can be divided in 4 main types:

- Force/torque and tactile sensors.
- Proximity or distance sensors.
- Vision systems.
- Etc.



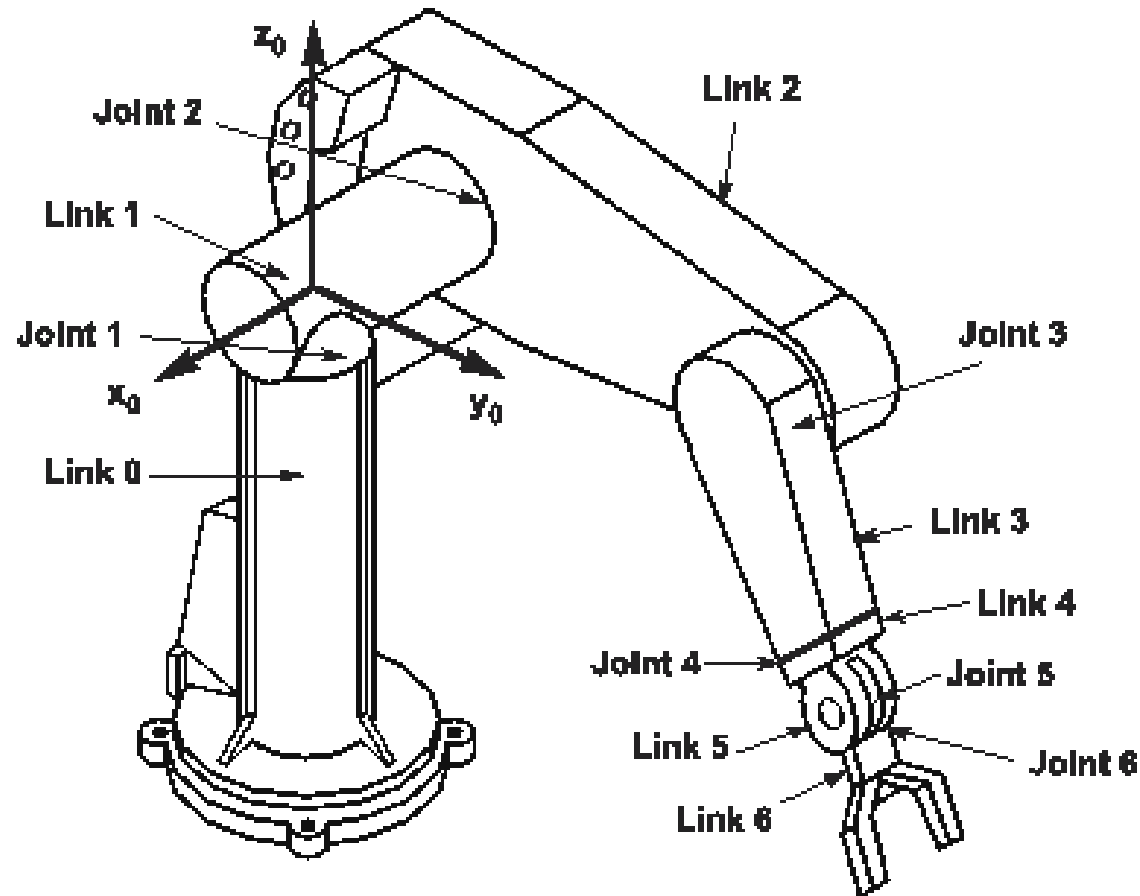
Elements of a Robot

Type of sensor	Main function	Examples
Motion and position	Movement	Encoders, Gyros
Proximity	Avoidance	IR, Ultrasonic
Vision and camera	Seeing	2D or Depth Cameras
Force and torque	Pressure	Force or torque sensors
Range	Distance	LiDAR, Ultrasonic
Tactile and touch	Touch	Pads, Artificial skin
Environmental	Safety	Temp, Gas, Humidity

Type of robots: Robotic Arms

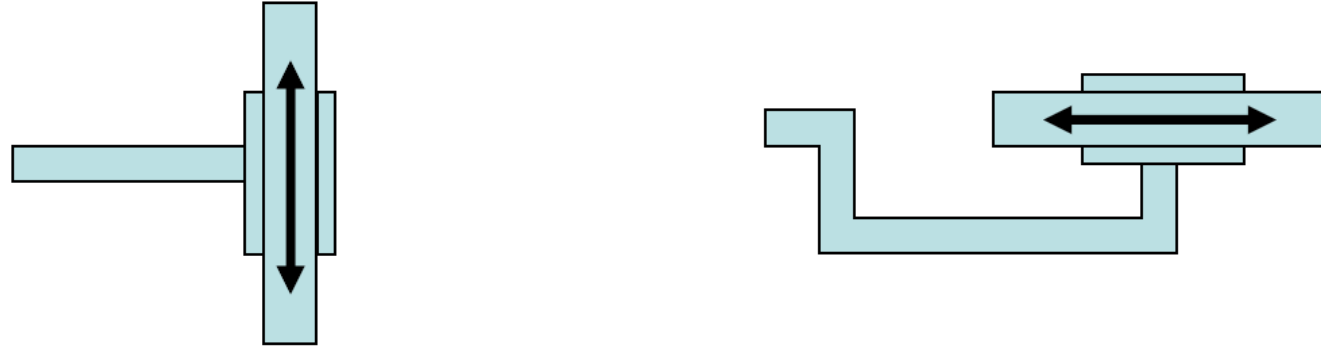
Include:

- **Links:** rigid interconnected bodies.
- **Joints:** can be **prismatic** and **revolute** (rotational) joints.

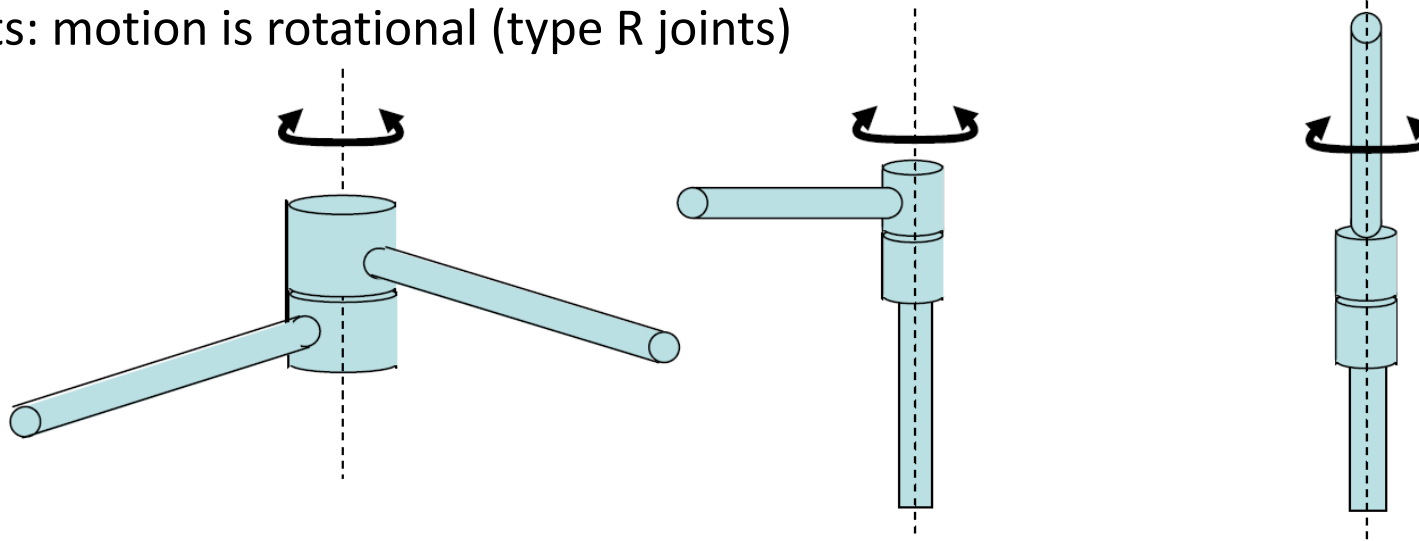


Type of robots: Robotic Arms

- Prismatic joints: motion is translational (type T joints)



- Revolute joints: motion is rotational (type R joints)



- Other types of joints (e.g., spherical, helical) can be described as combinations of R and T joints

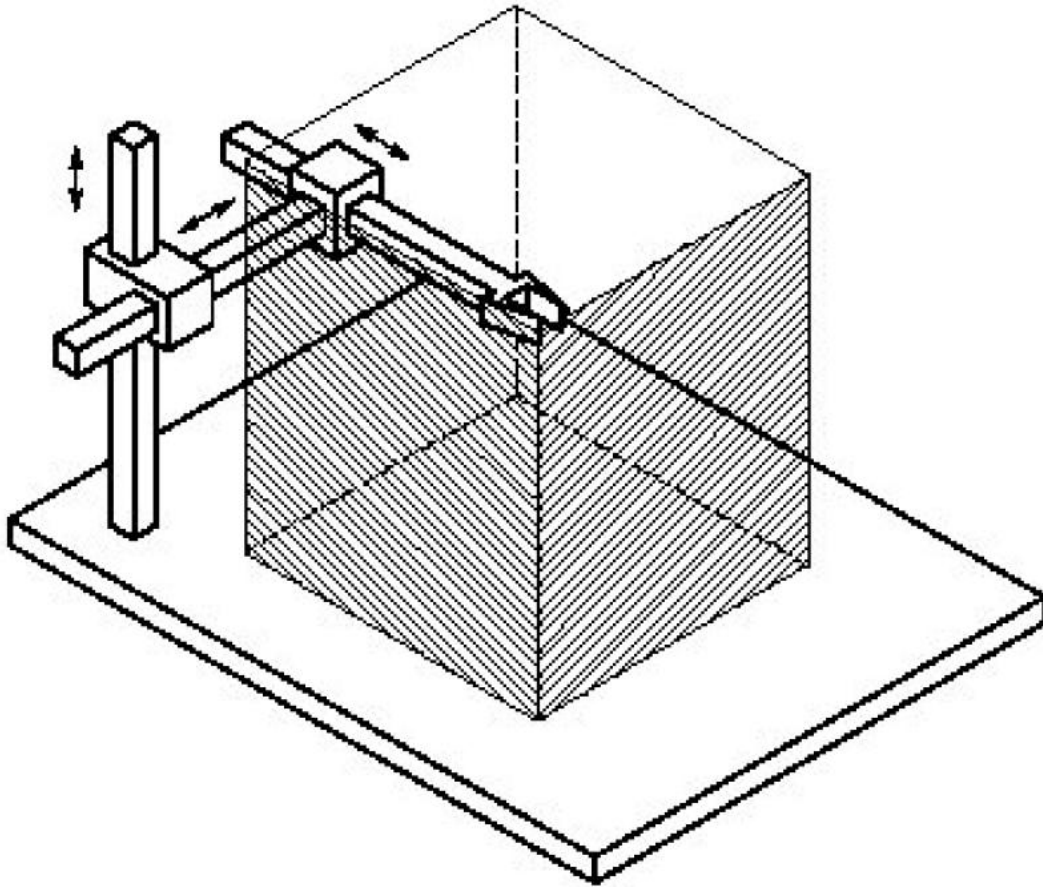
Type of robots: Robotic Arms

Robotic arms are divided into three main families:

1. Serial.
2. Parallel.
3. Hybrid.

Serial robots: Cartesian - TTT

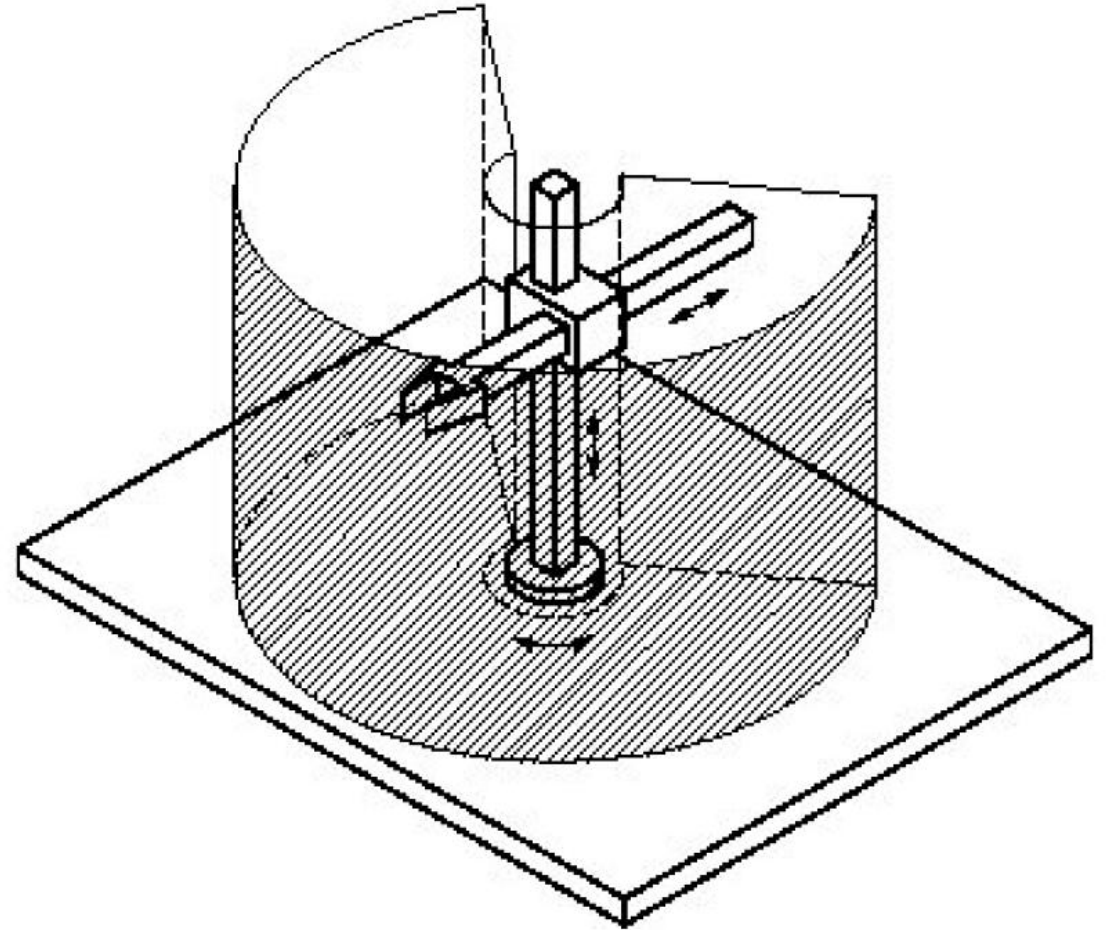
Typically used for palletizing heavy loads



Serial robots: Cylindrical – RTT

Applications:

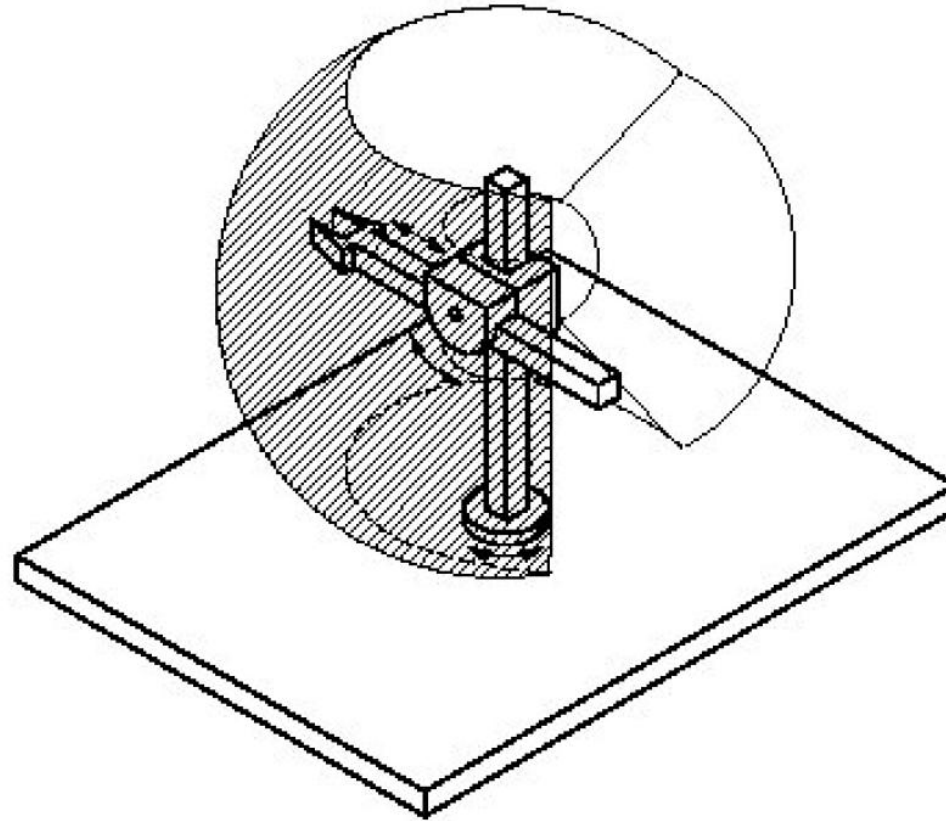
- Electronic component fastening
- Assembly operations



Serial robots: Spherical – RRT

Applications:

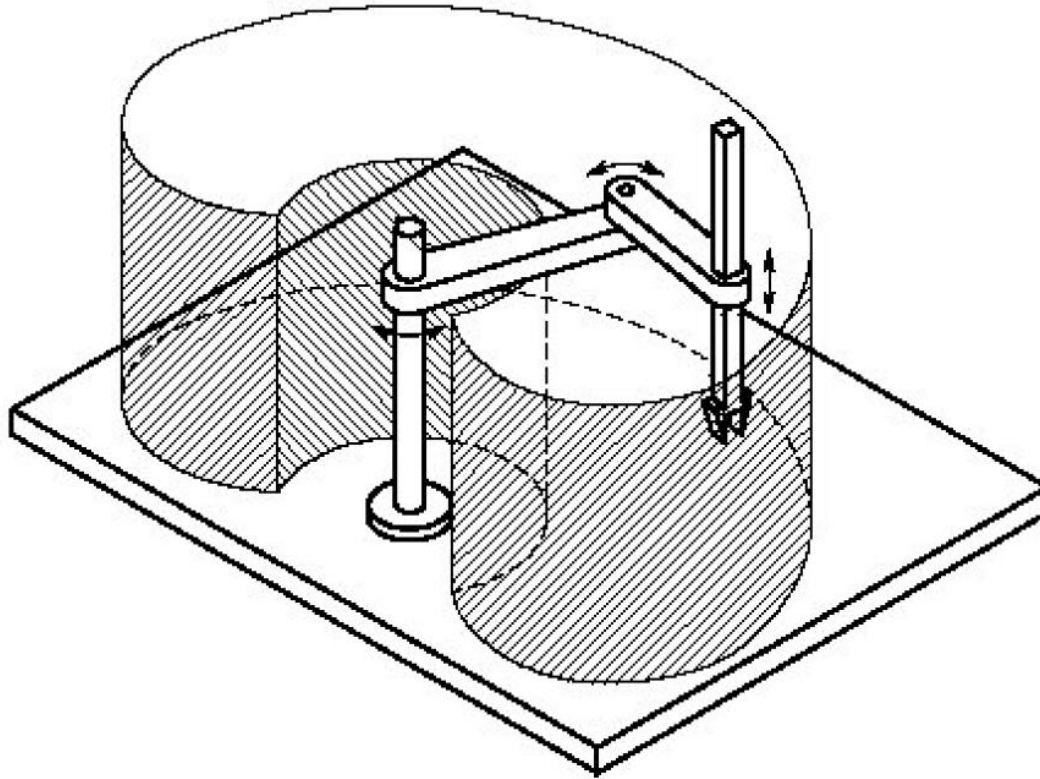
- Handling operations.
- Assembly tasks.
- Welding and painting.



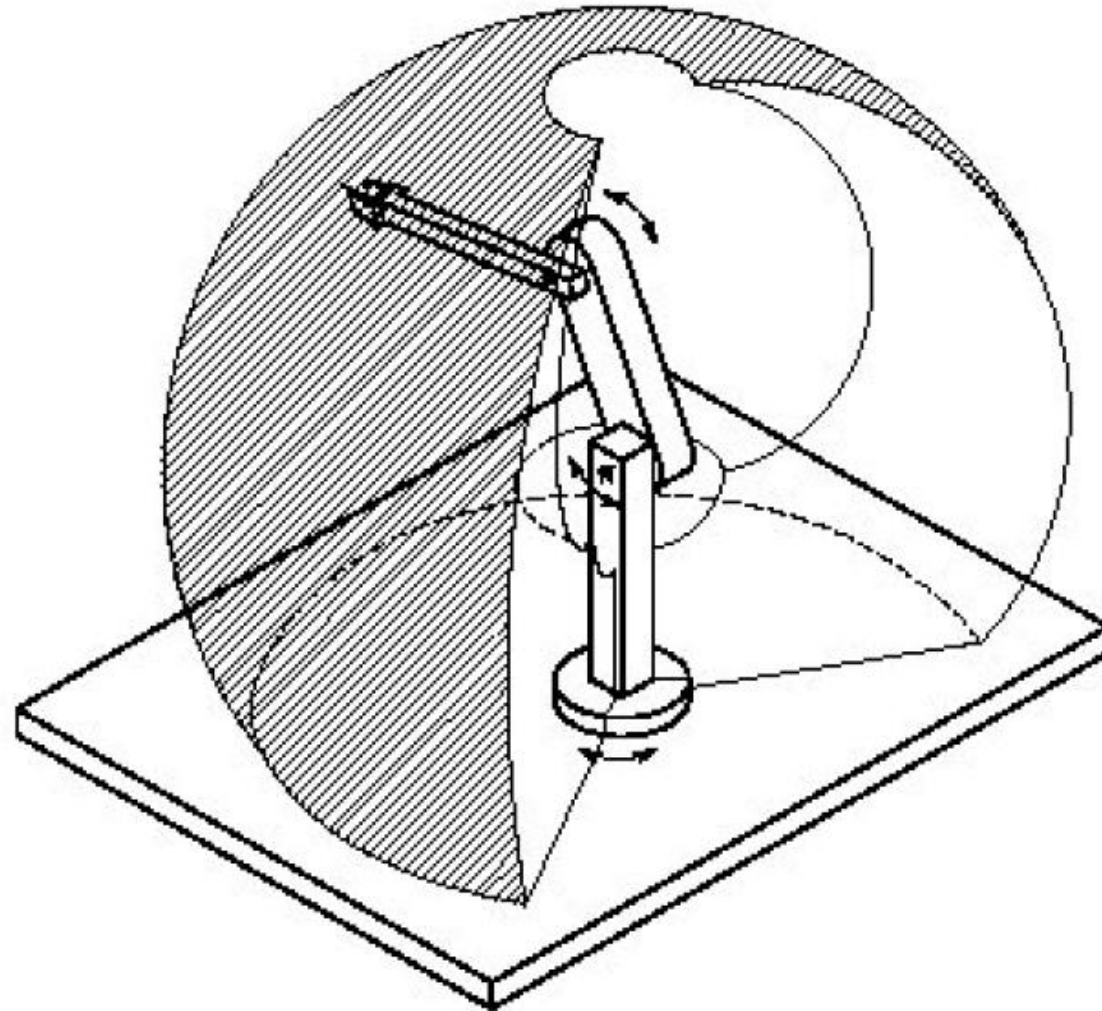
Serial robots: SCARA -RRT

Selective Compliance Assembly Robot Arm.

Main application area:
Palletizing or depalletizing.

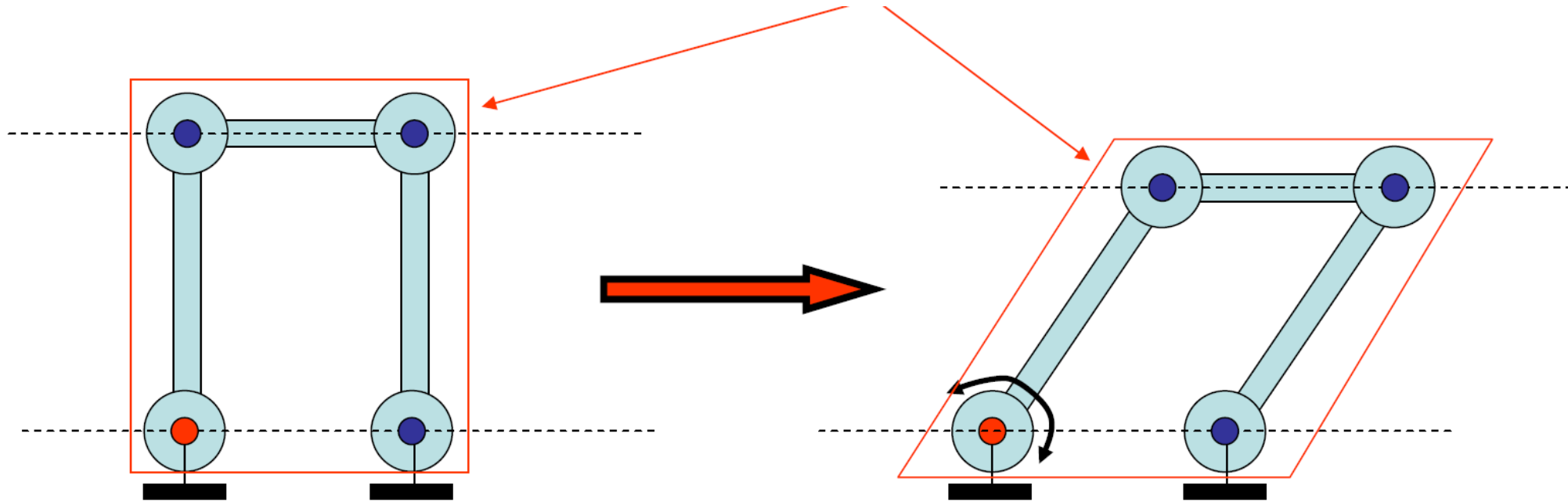


Serial robots: Anthropomorphic – RRR



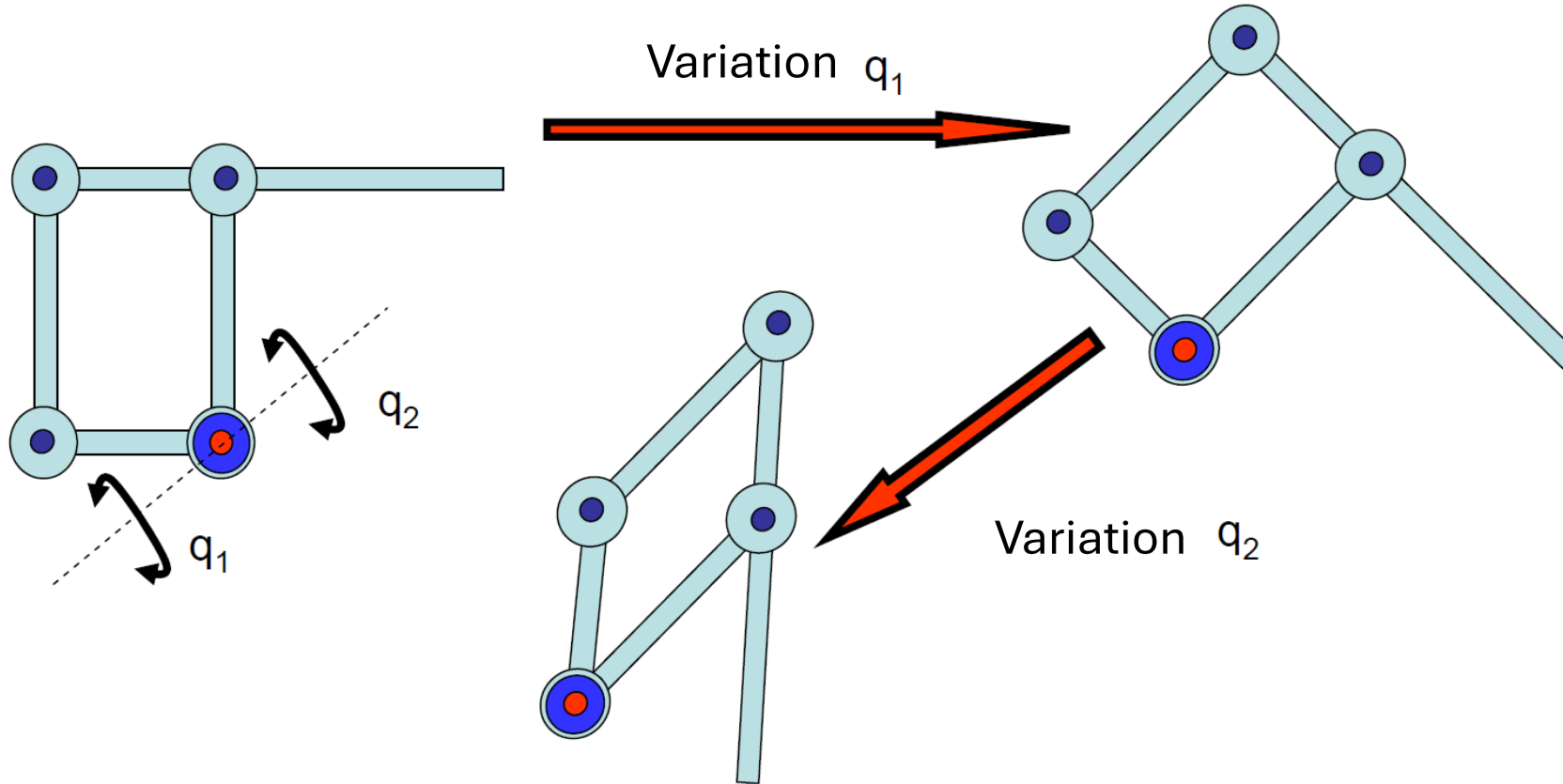
Parallel Robots: 4-Bar Mechanism

Parallel robots are also known as **closed-kinematic-chain** robots.



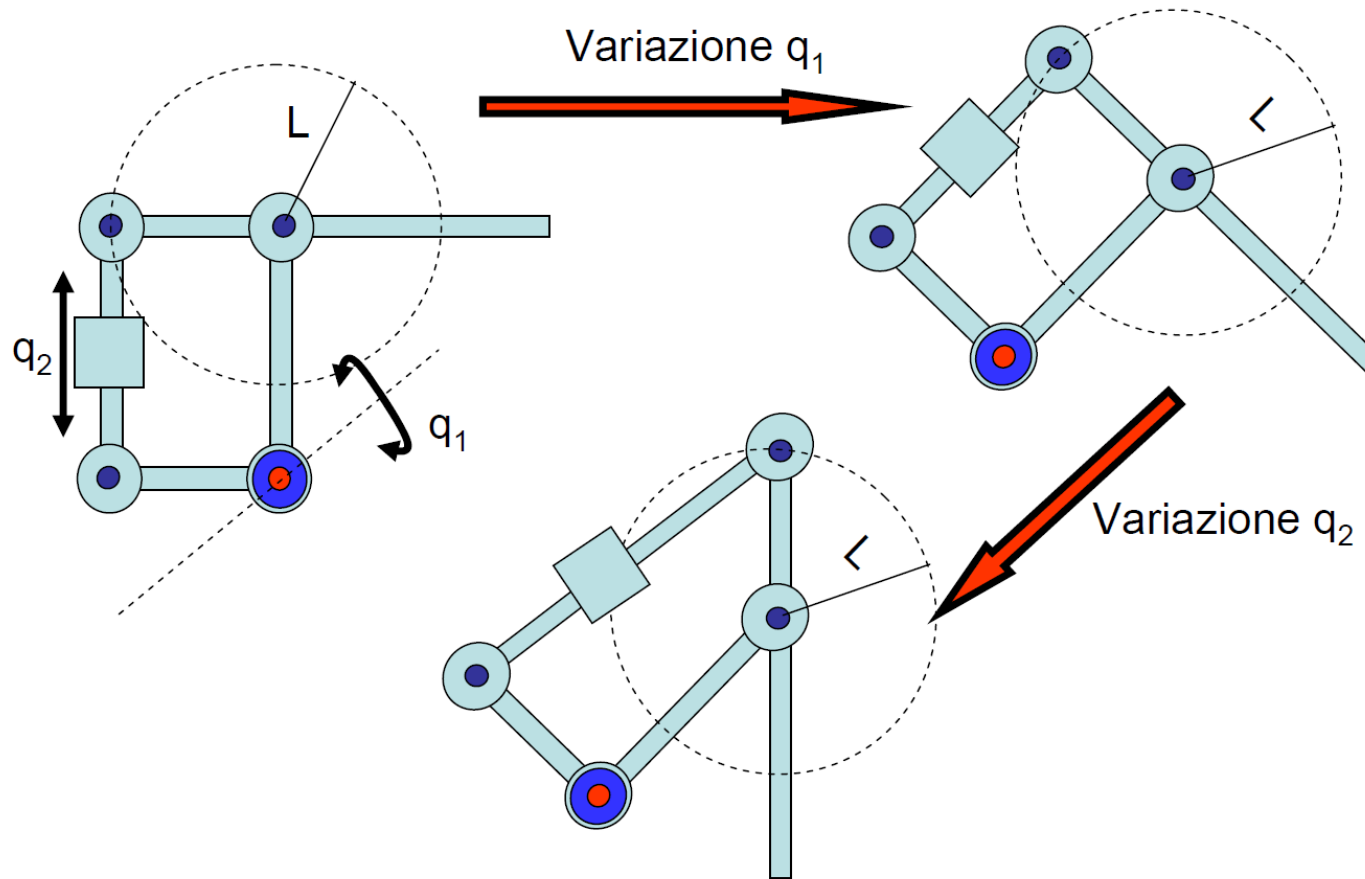
- The 4-bar structure is usually **driven by a single motor**.
- **Main features:**
 - The upper link remains parallel to the base during motion.
 - High rigidity (greater than that of serial chains).

Parallel Robots: 5-Bar R



- The 5-bar structure is usually **driven by two motors**, often **coaxial**.
- **Main feature:**
 - Possible configurations also include the upper link **not parallel** to the base.

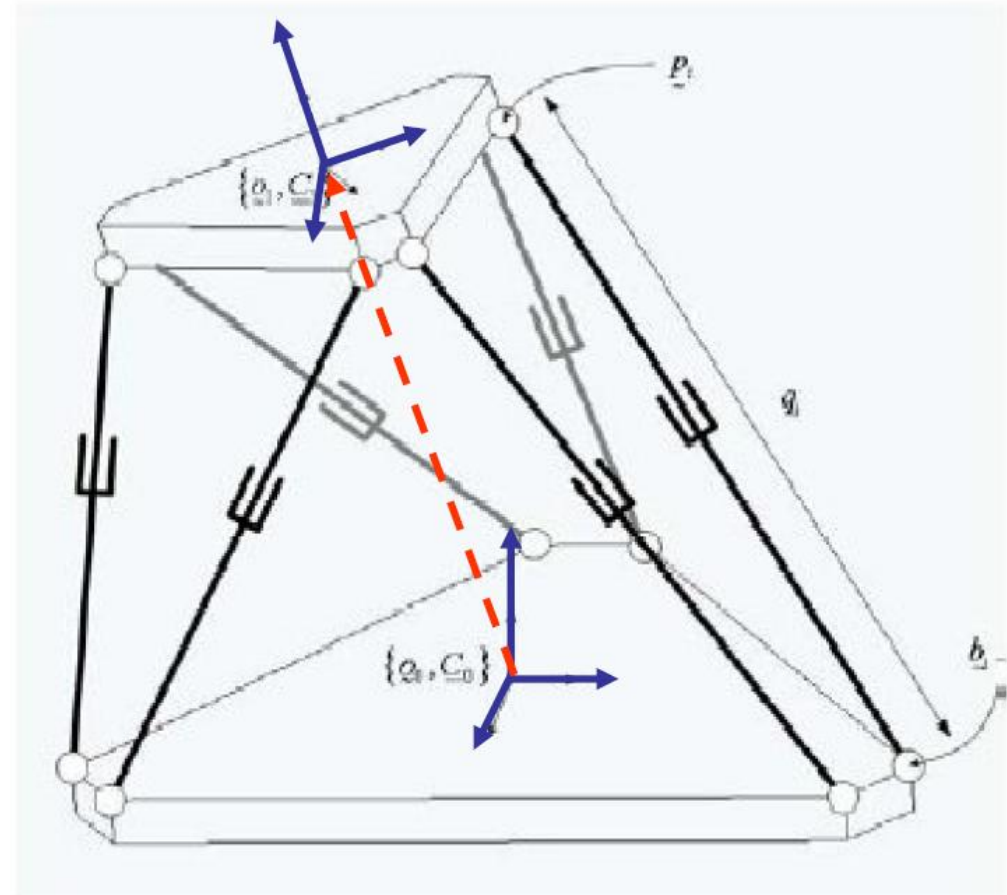
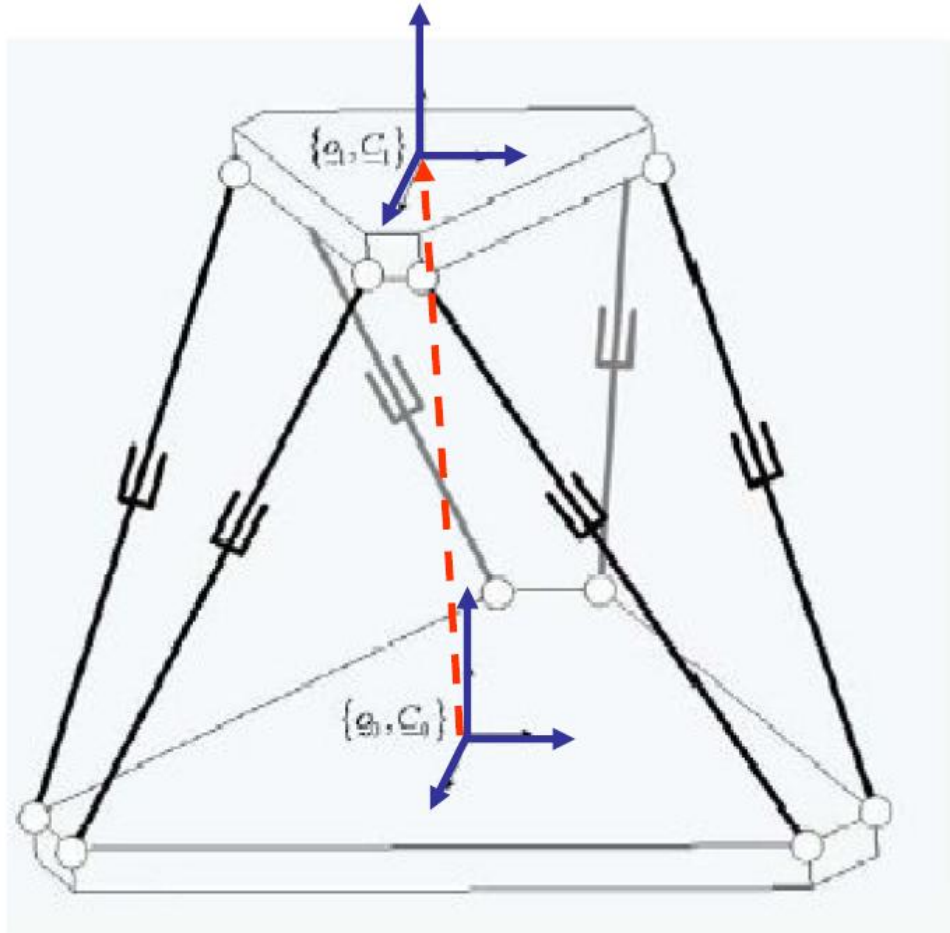
Parallel Robots: 5-Bar Hybrid (RT)



- Combines **revolute (R)** and **prismatic (T)** joints
- Offers the **same kinematic characteristics** as the pure 5-bar mechanism
- Enables both **rotational (q_1)** and **translational (q_2) motion** in the end-effector

Parallel Robots: 6-DOF

The motion of the end-effector is no longer planar but follows a **3D trajectory**.
So, the end-effector position is defined by **six coordinates** (3 positional and 3 orientational variables).
These are highly complex mechanisms.

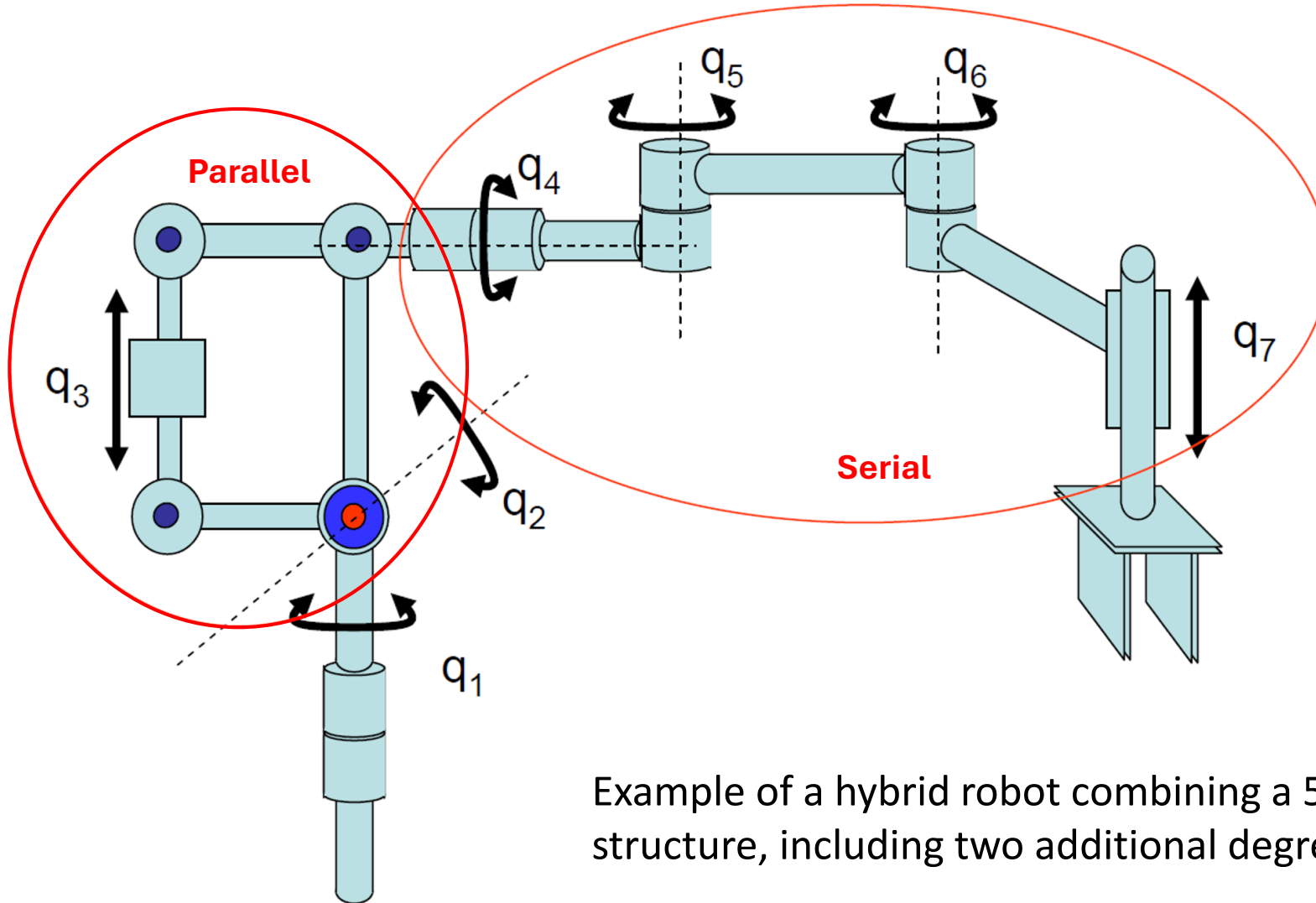


Parallel Kinematics



Hybrid Robots

Fusion of serial robots and parallel robots.



Example of a hybrid robot combining a 5-bar mechanism with a SCARA structure, including two additional degrees of freedom (q_1 and q_4).

Type of robots: Vehicles

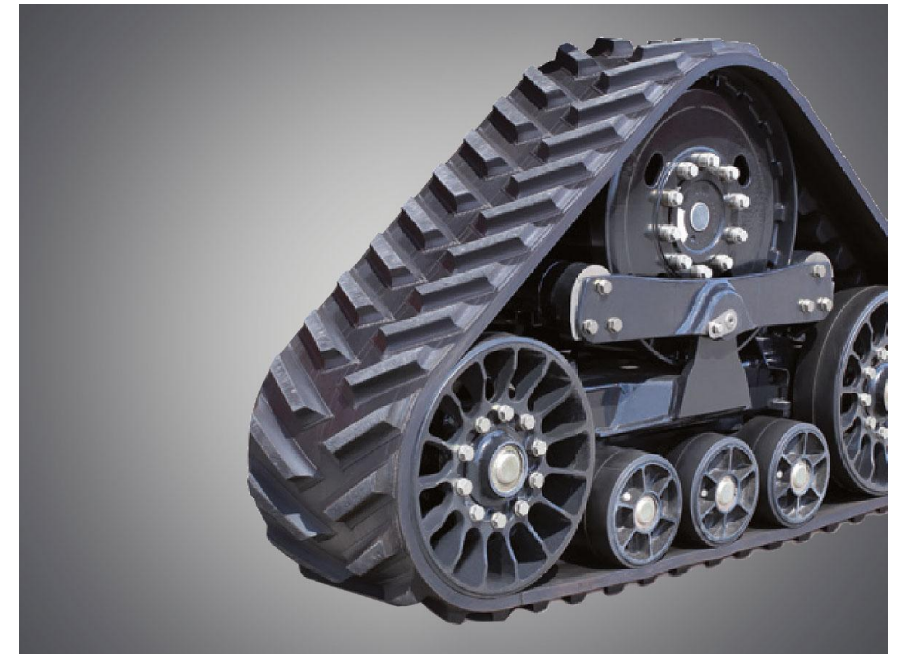
Vehicles can achieve locomotion through:

- Wheels.
- Tracks.
- Legs.



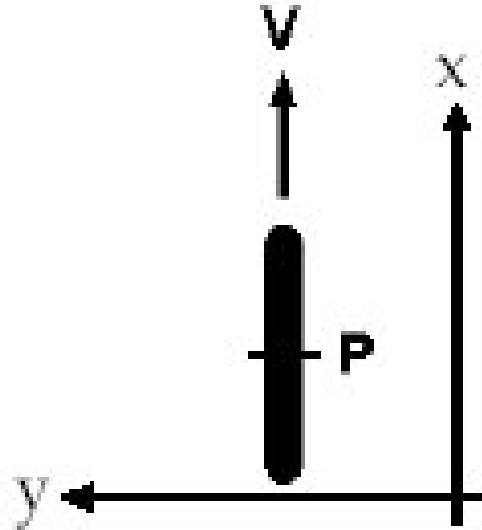
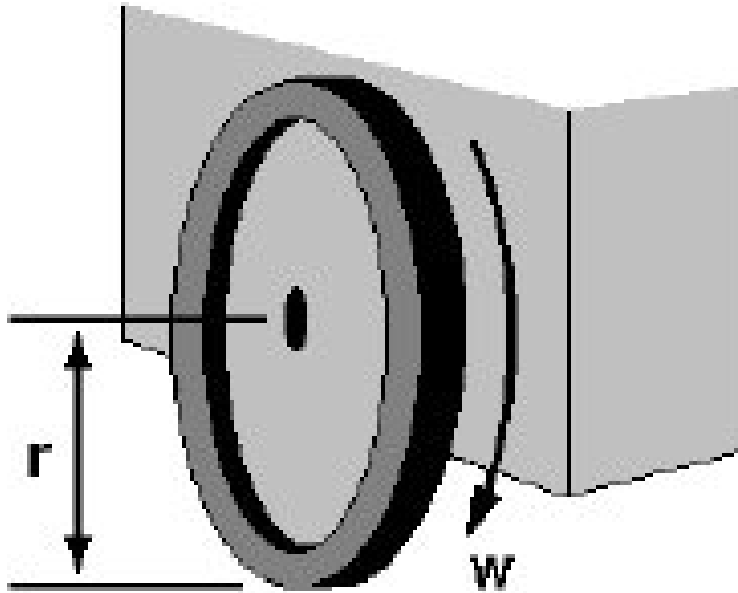
Several families of vehicles:

- Unicycle models.
- Car-like vehicles.
- Skid-steering vehicles.
- Omnidirectional platforms.



Wheel-based drive systems

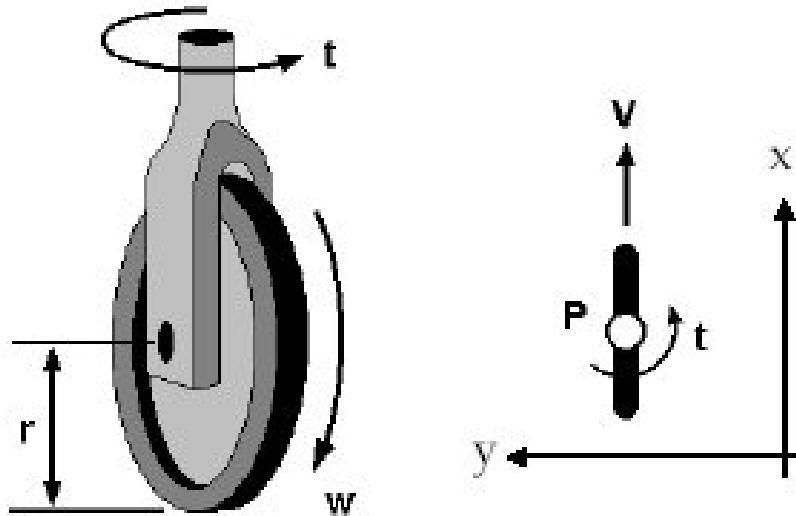
Fixed Wheel



Point P cannot move (unless it slips) in the direction of the wheel's rotational axis.
It has **Nonholonomic** constraint.

Wheel-based drive systems

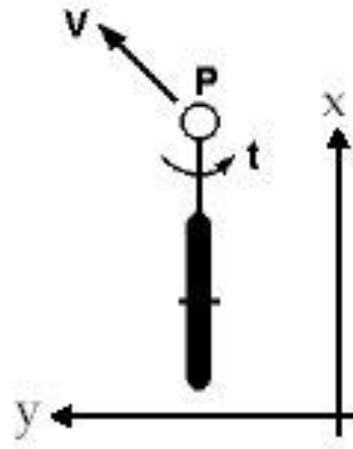
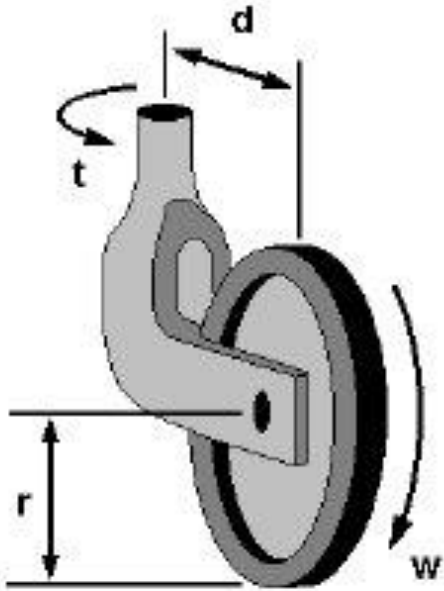
Steerable Centered Wheel:



Restriction on the robot's mobility.

Wheel-based drive systems

Eccentric Steerable Wheel (Castor Wheel):



- w : wheel rolling rotation.
- r : wheel radius.
- d : offset distance of the swivel from the wheel center.
- t : steering angle or swivel rotation (castor rotation).

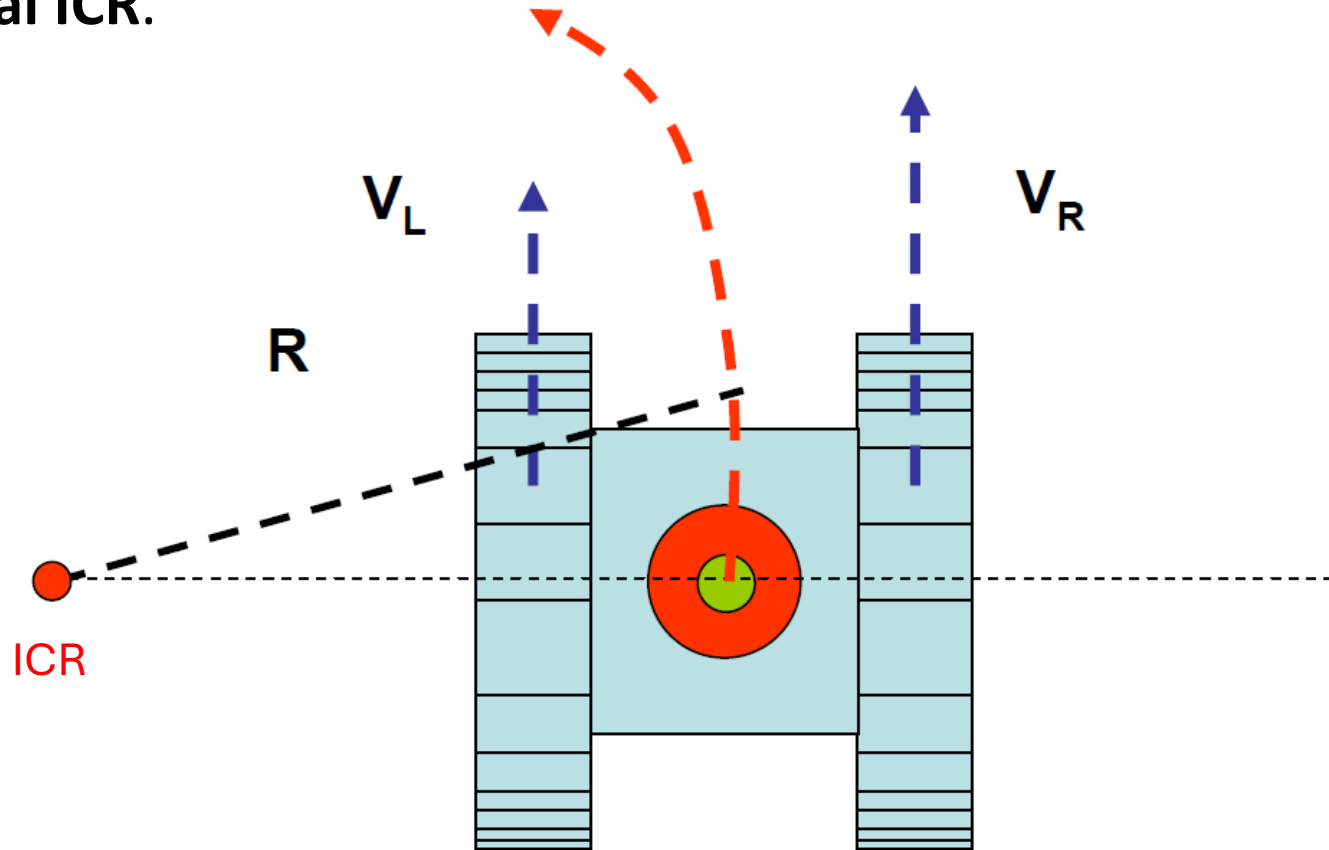
Omnidirectional property.

Tracked propulsion systems

Straight-line motion: tracks move at the same speed and in the same direction.

Rototranslational motion: - with **external ICR** (Instantaneous Center of Rotation).
- with **ICR on the axis of a track**.

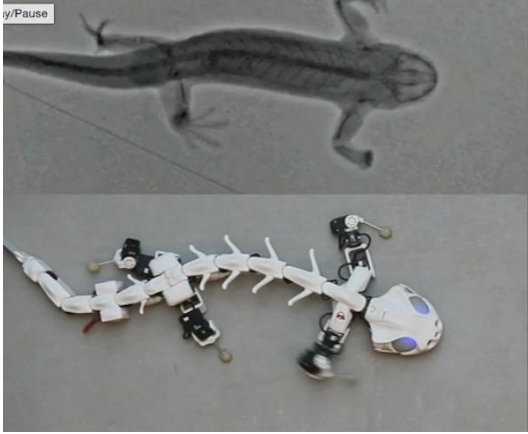
Rotational motion: with **internal ICR**.



Point **P** cannot move (unless it slips) in the direction of the wheel's rotational axis.

Legged locomotion mechanisms

Ability to easily overcome obstacles.



Can have very complex structures.

